



Module: Electronics, system components

Series of TD n° 6

Exercise 1:

- 1) What are the advantages of computer networks?
- 2) Several criteria make it possible to classify networks? What are the different networks classified according to geographic coverage (distance) and architecture?
- 3) Name the hardware components that can be found in a network?
- 4) Indicate the type of network (LAN, MAN, WAN or PAN) to use for each of the following cases:

Description	Network
A network that connects computers in a building.	
A network that connects staff devices (laptop, smartphone, etc.).	
A network that connects a company's subsidiaries internationally.	
A network that connects the subsidiaries of a company located in a city.	

Exercise 2:

- 1) What element is it?
 - a) Its role is to pass packets from one network to another, at best, according to a set of rules.
 - b) Is a hardware element having a certain number of ports? Its sole purpose is to recover binary data arriving on a port and broadcast it to all ports.
 - c) Is a hardware element having a certain number of ports, it analyzes the frames arriving on its input ports and filters the data in order to direct it only to the appropriate ports.
- 2) Either the following computer equipment:
 - a) What must be done so that all three computers can print to the same printer?
 - b) Diagram your solution in the figure above.



Exercise 3: Choose the right answer:

- 1.** Network devices link together using a variety of connections like:
 - a. wood cabling
 - b. Copper cabling
 - c. Plastic cabling
- 2.** Benefits from networking include:
 - a. Higher cost licensing
 - b. Centralized administration
 - c. Decreased communication capabilities
 - d. Assuing file duplication and corruption
- 3.** A group of interconnected computers under one administrative control group that governs the security and access control policies that are in force on the network.
 - a. LAN
 - b. WAN
 - c. MAN
 - d. PAN
- 4.** A group of wireless devices that connect to access points within a specified area. Access points are typically connected to the network using copper cabling.
 - a. LAN
 - b. MAN
 - c. PAN
 - d. WLAN
- 5.** Network that connects devices, such as mice, keyboards, printers, smartphones, and tablets within the range of an individual person. They are most often connected with Bluetooth technology.
 - a. LAN
 - b. MAN
 - c. WAN
 - d. PAN
- 6.** Network that spans across a large campus or a city. Consisting of various buildings interconnected through wireless or fiber optic backbones.
 - a. PAN
 - b. LAN
 - c. MAN
 - d. WLAN
- 7.** Connections of multiple smaller networks such as LANs that are in geographically separated locations. The most common example of a WAN is the Internet.
 - a. LAN
 - b. PAN
 - c. WLAN
 - d. WAN
- 8.** Devices which are connected directly to each other without any additional controlling devices between them. Each device has equivalent capabilities and responsibilities.
 - a. Client/server networks
 - b. Peer-to-peer networks
 - c. WAN
 - d. MAN

- 9.** In a client/server model, the client requests information or services from the server. The server provides the requested information or service to the client.
- a. MAN
 - b. Client/server networks
 - c. WAN
 - d. Peer-to-peer networks
- 10.** An IP address is a unique number that is used to identify a network device and is represented as a ----- bits binary number, divided into ----- octets (groups of eight bits):
- a. 16 , 8
 - b. 64 , 4
 - c. 32 , 8
 - d. 32 , 4
- 11.** Convert 11111101 to decimal
- a. 255
 - b. 253
 - c. 251
 - d. 254
- 12.** Convert 24 to binary
- a. 00011000
 - b. 11110000
 - c. 00010100
 - d. 00000110
- 13.** To determine the IP address for a certain computer using CMD tool , you type the command:
- a. ip config
 - b. ip address
 - c. ipconfig
 - d. ipbinary

Good luck
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